Lecture 02 — Vector arithmetic and geometry

Wed Aug 26

Last updated: August 18, 2026.

Reading: Boyd 1.2–1.3. Related objectives: Lectures. Schedule.

Learning objectives

By the end of class, students should be able to:

1. Add, subtract, and scale vectors.
2. Interpret those operations geometrically.
3. Plot vectors in $\mathbb{R}^2$ and recognize corresponding operations in $\mathbb{R}^3$.
4. Check whether an answer has sensible dimensions and units.

Fill in

If a and b are n-vectors, the sum $a + b$ is the n-vector whose ith entry is ________________. Addition is only defined when _________________.

The identity for addition is _________: $a + 0 = a$. The difference $a - b$ is ______________.

If β is a scalar, the scalar–vector product βa is the n-vector whose ith entry is ______________. In particular, $(-1)a$ is written _______.

Geometry in $\mathbb{R}^2$: if a is drawn as an arrow, βa with $\beta > 0$ points ____________ and has length scaled by ________. If $\beta < 0$, the arrow points ______________.

Adding a scalar to a vector is ______________ in the mathematics of this course (even if some software will do it). Concatenating $(1, 2)$ with $(3, 4)$ is also ______________ vector addition.

Worked in class

W1. A point on the quad is $p = (2, 1)$ (east, north, in 10 m units). A walk is the displacement $a = (3, -1)$. Compute $p + a$ and sketch p, a, and $p + a$ in $\mathbb{R}^2$. Then compute $a - p$. Does $a - p$ have the same units as p?

W2. Two rain gauges, in inches, over three days:

$$r = (0.40, 0, 1.10), \qquad s = (0.15, 0.20, 0.30).$$

Compute $r + s$ (combined rainfall if you could pour both gauges into one story). Compute $2r$. Why is $r + (1, 1)$ not a valid vector sum here?

W3. Sanity check: can you form $(4, 1) + (0, 2, 5)$? Can you form $(18 \text{ cm}, 4 \text{ mm}) + (1, 1)$ as a meaningful measurement vector?

Your turn

Y1. Compute

$$\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \qquad \text{and} \qquad (-2) \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}.$$

Y2. In $\mathbb{R}^2$, sketch $u = (3, 1)$ and $\frac{1}{2}u$. In words, what did scaling by $1/2$ do? What would $-u$ look like?

Y3. A 2-vector of temperatures in °C is $x = (10, 14)$. A classmate types $x + 5$ in software and gets $(15, 19)$. Is that the same as a vector sum $x + v$ for some 2-vector v? If you need "five degrees warmer each day," what vector should you add?